

ENHANCING DEMENTIA DETECTION: AUGMENTING THE FACE MESH

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ABSTRACT

Alzheimer's disease (AD) stands as a prevalent form of dementia, primarily impacting the elderly, and presently lacking any known cure. Early screening and intervention have demonstrated efficacy for both current and prospective treatments. With the global population aging, early detection of AD has become imperative. Language-based approaches for dementia detection have received considerable attention in research due to dementia's nature of hurting cognitive and language functions. Facial features, on the other hand, have been comparatively neglected despite their potential in identifying AD, due to a lack of usable dataset. AD patients show different patterns while displaying facial expressions because of Apathy, a phenomenon common in AD patients. This makes facial features non-negligible in dementia detection. The face mesh, comprising 478 landmarks, serves as a rich and informative facial representation. This study concentrates on leveraging face mesh data for AD detection. We explored data augmentation techniques to address the challenges of over-fitting, particularly impactful in training neural networks on a small dataset. Deep learning methods, including transformer, Convolutional Neural Networks (CNN), and Long Short-Term Memory (LSTM) networks, were employed to classify dementia patients and healthy subjects. By augmenting the dataset by rotation, shift, and scaling, we improved the performance of the deep learning methods. The combined use of 1D CNN networks and data augmentation method reported the highest accuracy of 76%. These results confirmed the effectiveness of our data augmentation methods. They also underscore the potential of face-related features in enhancing early detection approaches for AD.

Index Terms— data augmentation, dementia detection, deep learning, face mesh

1. INTRODUCTION

Dementia, a degenerative neurological disorder, progressively hampers an individual's cognitive capacities and day-to-day functionality. Primarily affecting the elderly, dementia afflicts approximately 55 million people globally, with an additional 10 million new cases surfacing each year [1]. Patients grappling with dementia may manifest symptoms like memory loss, impaired judgment, and bewilderment, encountering challenges in articulation and comprehension. Some may even find themselves disoriented in familiar surroundings. As the illness advances, those afflicted may also display behavioral and psychological manifestations, including but not limited to depression, anxiety, agitation, and aggression [2].

The widespread application of conventional methods for diagnosing dementia, such as neuroimaging and psychological testing, is severely hampered. Firstly, these methods need a lot of resources and money. The conventional methods involve costly medical infrastructure, skilled professionals, and a large amount of time commitment.

Therefore, it is difficult for these diagnostic processes to scale up to a wider population. Furthermore, people may be discouraged from willingly engaging in these assessments due to the invasive nature of neuroimaging and the complexity of psychological examinations, which might lead to poor acceptance rates and possible delays in the diagnosis of dementia. A low-cost and scalable screening method would be highly favored to recognize possible biological indicators for dementia, especially in the early stage. Contrary to the huge amount of studies [3, 4, 5, 6, 7, 8] to search dementia indicators from speech or transcriptions, facial features have not gotten as much attention, despite being demonstrated to be useful indicators [9]. Some researches [9, 10] have investigated the use of facial features or face images for dementia detection; however, these studies utilized very small and restricted datasets, which may introduce institutional bias into the results, yielding a high performance with no generalization ability when applied to other data.

Motivated by the under-researched condition in the area of using facial features to detect dementia, we aimed to investigate applying facial features to detect dementia. In this study, we especially explored the use of face mesh. We proposed a data augmentation method to mitigate the lack of data and investigated three common deep learning models to improve the classification performance.

The structure of the paper is organized as follows: Section II introduces the related works that used facial features to detect dementia. Section III describes the dataset, face mesh, and proposed methods used in this research. Section IV presents the experimental settings as well as the results and performance of the proposed methods. Section V concludes this paper.

2. RELATED WORKS

Asplund *et al.* [11, 12] reported notable variations in facial expressions of people with dementia and showed it possible to identify dementia based on the variations in facial expressions. According to Asplund *et al.* [11], certain dementia patients have a tendency to show fewer facial expressions. Their investigation, which was done under both happy and unpleasant stimulus settings utilizing the Facial Action Coding System (FACS), revealed a decreased capacity for expression production. They investigated two techniques for reading facial expressions in dementia patients and found differences between them in a follow-up research [12]. Patients with dementia have been seen to display reduced facial cues in quantity or clarity. These results imply that variations in the facial expressions of those suffering from dementia may act as markers for dementia.

In recent years, several studies [9, 10, 13] have gathered facial expression data from individuals with dementia, exploring the potential of utilizing facial expressions for dementia detection. To overcome the under-researched aspect of facial expression in dementia studies relative to speech and language data, the researchers presented an automated method for dementia identification from a

human face. They retrieved face-related features from spoken dialogues involving 24 people and attempted to categorize dementia. To categorize individuals into two groups, the combination of L1 logistic regression and face-related features produced a detection performance of 0.82 on accuracy. Lip activity, action units (AUs), and eye gaze were among the most contributing features that were found. Based on a spoken dialog system, the results showed that the system can identify dementia and provide help to medical professionals.

Umeda *et al.* [10] aimed to explore whether artificial intelligence could distinguish the faces of individuals with cognitive impairment from those without. Using samples collected from 121 patients with cognitive impairment and 117 cognitively sound participants, they tested five deep learning models. Xception with the Adam optimizer performed the best, showing 87.31% sensitivity, 94.57% specificity, and 92.56% accuracy. The study revealed a significant correlation between the “Face AI score” and cognitive impairment, suggesting the potential use of AI for developing a facial biomarker for dementia. Even though their study produced excellent accuracy results, it is vital to remember that they used raw face images, and institutional bias was a point of worry mentioned in [10]. The machine learning algorithm might be misled by institutional bias and end up learning about the environment instead of dementia markers. Zheng *et al.* [14] explored the use of facial features for dementia detection. They extracted face mesh, AUs, and HoG (Histogram of Oriented Gradients) features from interview videos involving dementia patients and healthy subjects. Utilizing a voting system in conjunction with SVM as the classifier, they attained a 73% accuracy, specifically leveraging the AUs data.

3. METHODS

3.1. PROMPT Dataset

Approved by Keio University School of Medicine Ethics Committee, the Keio Medical School collected voice and video data from participants of healthy subjects and dementia patients for the PROMPT dataset [15] with informed consent obtained. In the collection process, the dementia patients participated in several interview sessions conducted by professionals, with at least a 2-week gap between sessions. During each session, the patients were required to undergo several interviews. The video and audio recordings of the sessions were captured separately from two microphones and an Intel Realsense camera. One microphone was set at the patients’ place and another set at the doctor’s place. The camera was set on the table in front of the patients, with a distance of about 1 meter between them. The study involved three tasks: free speech, where participants engaged in casual guided conversations with the interviewer; question and answer (Q&A), where participants responded to complex situational queries. Additionally, participants described the content of an image depicting a scenario with children arguing over food in the picture description task.

The PROMPT dataset comprises 446 samples collected from dementia patients and healthy subjects, encompassing both dementia and healthy cases, obtained from 117 subjects, each sample spanning approximately 10 minutes. Despite its richness, this dataset falls short in size compared to conventional machine learning datasets, posing challenges in training a generalizable and robust deep learning model. To address this limitation, our research introduces a novel approach by leveraging data augmentation. This strategy aims to overcome the over-fitting problems inherent in smaller datasets, enhancing the model’s ability to generalize effectively and to facilitate the training of more proficient deep learning models.

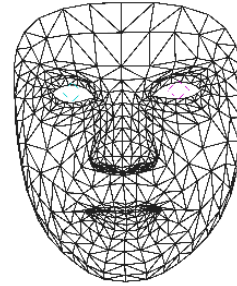


Fig. 1. An illustration of a face mesh generated using MediaPipe. The blue and red strokes annotate the position of the pupils.

3.2. Face Mesh

Face meshes can be represented as a set of points or vertices that define the facial structure, rather than storing a complete image of the face. This reduces the risk of identifying individuals from facial features, thus enhancing anonymity and privacy. It also hides race, gender, or age information from the viewers. This function allows medical experts to watch and evaluate facial expressions and movements while maintaining patient privacy. This feature is especially helpful in telemedicine settings where patients are unable to see medical professionals in person. Because no third party will be able to determine who they are, it also motivates future AD sufferers to get screened early. Doctors and other medical professionals may evaluate patients’ status remotely while keeping secure the patients’ personal identification.

3.3. Data Augmentation

Although data augmentation has been extensively used in tasks dealing with image data or text data, data augmentation methods for face mesh data are not receiving attention from researchers. Face mesh data, as opposed to image data, employs a 3-dimensional structure as it contains x, y, z coordinates for each landmark it extracts. This is making the data augmentation different from that applied in the 2-dimensional images. Each face mesh can be transformed in 3-D space as long as the relative position among landmarks remains the same. In addition to that, this research aims to detect dementia by evaluating the temporal patterns from the face mesh time series data, which makes it necessary to keep temporal consistency before and after augmentation. In this research, analogous to the data augmentation methods commonly used for image data, we proposed 3 kinds of augmentation methods, featuring rotating, shifting, and scaling.

- **Rotation:** Randomly rotating the face mesh time series data. Specifically, for each face mesh time series data to be augmented, we randomly decided a rotation value (5 to 10 degrees) on each of the x, y, z axes. A rotation matrix is then created and applied to each landmark in each frame of the data.
- **Shifting:** Randomly shifting the face mesh time series data. Specifically, we randomly decide a shift between -0.5 to 0.5 on each of the x, y, z axes, which were later applied to each landmark of each frame, of the face mesh data.
- **Scaling:** Randomly scaling the face mesh time series data. Specifically, we first decide a factor between $[0.9, 1.1]$. Then this factor was applied to scale the face mesh data.

Fig. 2 showed examples of our data augmentation methods.

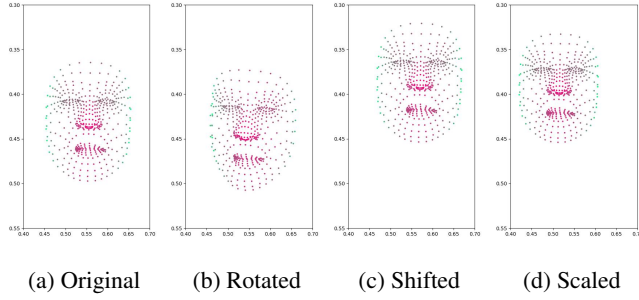


Fig. 2. Examples of augmented face mesh data by different methods. The color of the landmarks refers to the depth, with red ones closer to the camera and green ones further to the camera.

3.4. Models

We used Transformer, LSTM, and the 1D CNN network to perform classification between dementia and healthy samples. The transformer model contains only 1 transformer encoder layer, with 8 attention heads. The dropout rate was set to 0.5. The global average pooling layer compresses the hidden states of 32 to 1 and will be classified by the linear layer at the end of the model. We used the Adam optimizer with a learning rate of $2 \cdot 10^{-3}$ and set the batch size to 1024. The cross-entropy loss function was used to calculate the model loss.

The LSTM network is composed of 1 LSTM layer whose hidden state size was set to 64. The principal components are input into each cell of the LSTM layer and the hidden states of the last cell in the LSTM layer are forwarded to the linear layer (with 64 input features and 2 output features) and softmax layer for the final decision. We used Adam optimizer with a learning rate of $2 \cdot 10^{-3}$ and set the batch size to 256. The cross-entropy loss function was used to calculate the model loss.

The 1D CNN network is composed of 3 groups of CNN networks. Each group of networks is composed of a 1D CNN layer, a Max Pooling layer and a Relu activation layer. The 1D CNN layer has 32 in and out channels with a kernel size of 3. The Max Pooling layer has a kernel size of 2. A fully-connected network is finally applied to make predictions. We used Adam optimizer with a learning rate of $2 \cdot 10^{-3}$ and set the batch size to maximum. The cross-entropy loss function was used to calculate the model loss.

4. EXPERIMENT SETTINGS AND RESULTS

Following the common way to measure the performance of a machine learning task, we implemented a 5-fold cross-validation in the experiments. In each fold, 20% of the subjects were held out as the test set. In the training of each fold, we train transformer models, LSTM models, and 1D CNN models for 120 epochs, 400 epochs, and 120 epochs respectively. The epoch numbers are large enough for the model to converge on the training set, getting an accuracy of more than 99% on training data. We train the LSTM models for more epochs because they converge more slowly.

In this research, we implemented 3 kinds of data augmentation methods to increase the size of the data. Each time we applied 1 method of the three methods including rotating, shifting, and scaling. From each real face mesh sample video in the PROMPT dataset, we created an extra virtual sample by augmentation. As such, we created 3 extra datasets: virtual dataset 1 created by rotating, virtual

Table 1. The dementia detection accuracy performance of the different deep learning models with different data augmentation techniques.

Models	Baseline	Rotation	Shifting	Scaling	All
LSTM	0.698	0.705	0.705	0.707	0.636
Transformer	0.730	0.748	0.752	0.725	0.676
1D CNN	0.701	0.734	0.759	0.737	0.677

dataset 2 created by shifting, and virtual dataset 3 created by scaling. All of the 3 datasets share the same size as the real dataset. In the training, we used the real dataset with one of the 3 datasets, which increased the dataset size to double.

We implemented an incremental PCA method to reduce the dimensionality of the dataset with a batch size of 64 while processing dataset incrementally. The PCA is an unsupervised learning process, and it is performed on the augmented dataset (the real dataset with one of the virtual datasets) with each frame of the face mesh as a sample, this converts a frame of the face mesh into 32 principal components, which represent more than 99.9% variation information of the original face mesh dataset, according to our experiments. After doing the PCA reduction, a face mesh time series sample in the shape of $[N_{\text{steps}}, N_{\text{landmarks}}, N_{\text{axes}}]$ was converted to the shape of $[N_{\text{steps}}, N_{\text{components}}]$, where N_{steps} refers to the number of frames, $N_{\text{landmarks}}$ refers to the number of landmarks, N_{axes} refers to the number of elements each landmark has to annotate its location, and $N_{\text{components}}$ refers to the number of principal components.

Table 1 displays the highest accuracy performance of the 3 proposed data augmentation methods with different deep learning models employed, respectively. Column “All” refers to the performance when all 3 augmentation methods were employed. LSTM shows slight improvement in accuracy with each augmentation technique, reaching its peak accuracy with scaling (0.707). Transformer has the highest baseline accuracy (0.730). It performs best with shifting augmentation (0.752), but scaling reduces its accuracy to 0.725. The 1D CNN model demonstrates consistent improvement with all augmentation techniques, achieving the highest accuracy with shifting (0.759).

We noticed from the tables that shifting and rotation generally improve model performance, particularly for the Transformer and 1D CNN models. The augmentation of Mediapipe face mesh data shows promising results in enhancing the performance of deep learning models for dementia detection, with specific augmentations benefiting different models uniquely. Though applying a single data augmentation method on the deep learning models reports improvement on certain levels when all 3 methods were applied, the accuracy performance became worse. The reason behind it probably is when all 3 methods were used, the train data distribution was largely altered, which does not share commonness with test data anymore.

Traditional feature engineering and machine learning classifiers often outperform deep learning when data is limited. In this study, we implemented a traditional machine learning approach to classify dementia and healthy subjects. We processed face mesh time-series data into four representations: the original data, its differentiation, frequency spectrum, and spectrum differentiation. From these, we extracted statistical features such as mean, median, standard deviation, extremes, and key percentiles (e.g., 1%, 5%, 10%, etc.). Using a 5-fold cross-validation, we employed a Gradient Boosted Decision Tree (GBDT) classifier with 100 estimators, a learning rate of 1, and varying max depths. Table 2 shows the accuracy across different parameters and augmentation methods.

Table 2. Accuracy performance of Gradient Boosting Decision Tree when different augmentation methods were applied.

max depth	Baseline	Rotation	Shifting	Scaling	All
1	0.642	0.671	0.683	0.693	0.704
2	0.692	0.704	0.662	0.683	0.720
3	0.698	0.698	0.686	0.688	0.720
4	0.719	0.715	0.686	0.669	0.707

Unlike the LSTM and Transformer models, the GBDT model does not consistently improve with different augmentation. Its performance varies with different max depths. From Table 2, we see that a single data augmentation technique does not improve performance in most cases. However, it yields the highest accuracy of 0.720 when all the augmentation methods were applied, which is similar to the baseline model with a max depth of 4 (0.719). This inconsistency between feature engineering methods and deep learning methods was probably caused by the different processing before classifying. Data augmentation techniques such as rotation, shifting, and scaling primarily affect the raw data, adding variations to the original dataset. However, if feature engineering extracts aspects of the data that are invariant to these transformations or focuses on features that are not significantly altered by augmentation, the impact of augmentation might be minimized in the final model training. Compared with deep learning models that automatically find patterns in data, GBDT responds less to augmentation possibly due to the designed features not altering with augmentation.

5. CONCLUSION

By implementing and analyzing various models such as LSTM, Transformer, 1D CNN, and GBDT coupled with different data augmentation methods like rotation, shifting, and scaling, we have gained crucial insights into the potentials and limitations of deep learning and machine learning models in medical diagnostics. The proposed data augmentation methods improved the performance of different deep learning models. We achieved the best accuracy performance of 0.76 by employing a 1D CNN model with shifting augmentation. These results confirmed the effectiveness of our data augmentation methods. They also underscore the potential of face-related features in enhancing early detection approaches for AD.

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